Prosthetic limbs

A prosthetic limb is a device designed to replace a missing limb and help the user perform normal activities. Prostheses range from relatively simple mechanical devices to sophisticated electronic or robotic limbs that interact with the user's own nervous system.

NERVE SIGNALS FROM BRAIN TO ARM MUSCLES

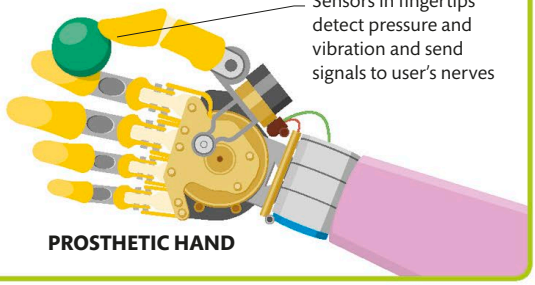
How a myoelectric lower-arm prosthesis works
Electrodes detect electrical signals from muscles in the residual arm. The signals are transmitted to a microprocessor, which converts them into data to instruct motors in the wrist and hand to move.

Prosthetic arms

The simplest prosthetic arms are mechanical, operated by cables running to the opposite shoulder and with a metal hook for gripping objects. More sophisticated myoelectric prostheses use electrodes to pick up muscle impulses from the remaining limb and convert these to electrical signals, which drive a motor to move the prosthetic arm and hand. For people who are missing most or all of their arm, targeted muscle reinnervation may be used. The nerve supply to a lost arm muscle is rerouted into a different muscle, such as a chest muscle; when the user thinks about moving the arm, the chest muscle contracts, and sensors placed over this muscle transmit signals to the prosthesis.

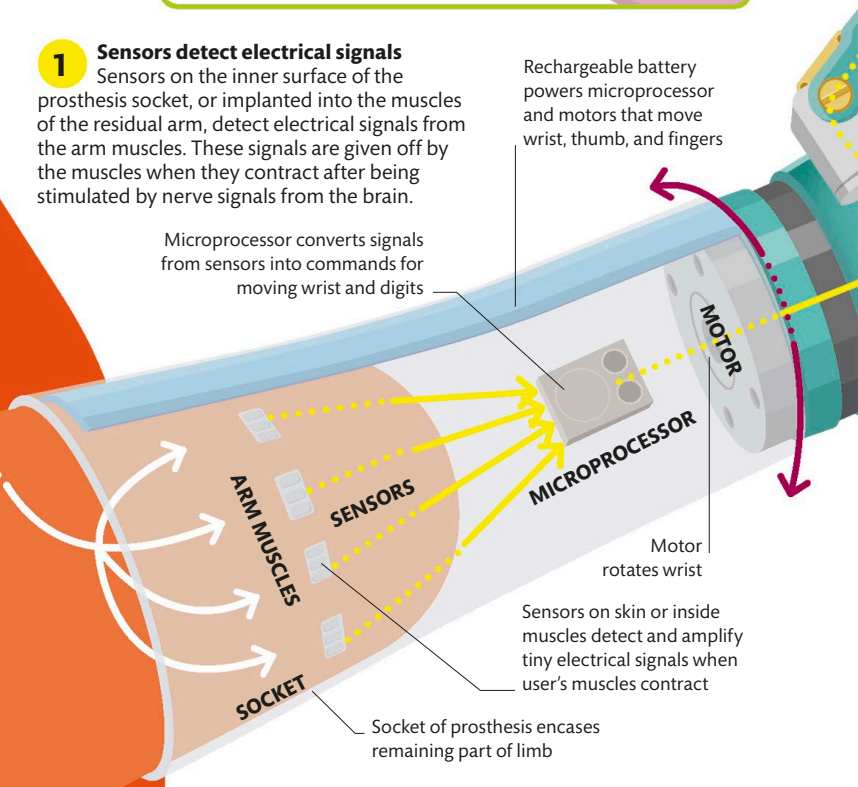
TOUCH SENSORS

Various prosthetic hands are being developed to restore a sense of touch to the user. These systems relay signals not just from the user's muscles to the prosthesis but from the prosthesis back to the brain. Sensors in the fingertips detect pressure or vibrations and relay this data to a computer chip. This converts the data into signals that are relayed to implants attached to nerves in the user's arm, which send impulses to the brain.



PROSTHETIC HAND

1 Sensors detect electrical signals
Sensors on the inner surface of the prosthesis socket, or implanted into the muscles of the residual arm, detect electrical signals from the arm muscles. These signals are given off by the muscles when they contract after being stimulated by nerve signals from the brain.



ATHLETES USING **RUNNING BLADES** HAVE TO **MOVE CONSTANTLY** TO STAY **BALANCED**

